INHA UNIVERSITY IN TASHKENT
School of Computers and Information Engineering
Spring semester 2025
Computer Networks

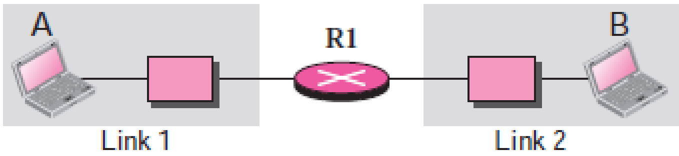
Name:- ______________________________ **ID:** __________________________ **Section:** ______

Submission Date: 24 /02/ 2025 (**23:00 Hr**)

NOTE :

1. Computer typed solution are not acceptable (**only handwritten** allowed)
2. Take the **print of this sheet** and solve on it.
3. Submissions not conforming to the given format will not be evaluated.
4. Submit on eclass, hardcopy submission not required.

Q1. Show the communication at the network layer in the below figure. Represent by drawing the network layer for the devices used in the communication.



Answer

Q2. Assume that each layer in a TCP/IP protocol suite adds a 20-byte header. A message of size 2000-byte is delivered through internet. Find the efficiency of the system. (efficiency is defined the ratio of the number of useful bytes to the number of total bytes)?

Answer

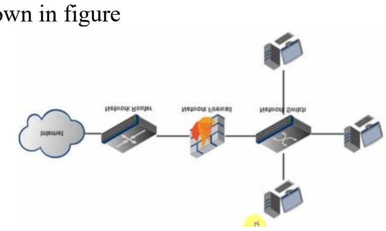
Q3. What header size (minimum) you will recommend at the transport layer of the TCP/IP protocol suite if a port number is defined using 16 bits (2 bytes). Give reason.

Answer

Q4. Why we need various addresses in computer networks to make communication possible. Also comment that why we do not need addresses at the physical layer for sending a message

Answer

Q5. You are appointed as a trainee in an organization. During the training, you have learnt to design a network using 3 computers, a switch, a firewall and a router. You proposed following n/w design shown in figure



Now you are assigned a task to design a network for the following scenario.

1. Use the current network as shown in rectangle, which consists of 3 desktop computers connected to 4-port switch.
2. Add 3 more laptops to this network
3. You are provided with one more **switch** and a **wireless access point**
4. Also connect this network to the outside world using internet.

Suggest a suitable Topology that can fulfil the above requirements

Answer